

RASHMI PUBUDITHA

AI/ML Engineer | Data Scientist

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SUMMARY

Machine Learning Engineer with hands-on experience building and integrating ML-powered features into production web systems using Python. Strong in applied machine learning (computer vision, NLP, time-series forecasting), API development (Flask/FastAPI), and deployment workflows with Docker, CI/CD, and AWS. Seeking ML Engineer roles focused on production ML and AI-driven products.

SKILLS

AI and Data science : LLMs, NLP, RAG, Generative AI, LLaMA, FAISS, OpenCV, NumPy, Pandas, Matplotlib, Pydantic, LangChain, Hugging Face, Statistics, Hadoop, Apache Spark, Image Processing, Jupyter Notebook

Machine Learning: TensorFlow, PyTorch, Scikit-learn, Model Development & Deployment, training & evaluation, feature engineering, time-series forecasting (SARIMA, LSTM), computer vision (YOLO, CNN)

Full Stack Development: Python, ReactJS, Flask, FastAPI, RESTful API, MySQL, MongoDB, Neo4j

Cloud/MLOps: Pytest, Docker, CI/CD, AWS (IAM, EC2, S3, RDS, SageMaker, Athena), Linux, Bash Scripting

Version Control: Git, GitHub, GitLab

Other Skills: Agile, Scrum, Fast Learning, Decision Making, Communication, Problem-Solving, Self Learning, Teamwork, Communication, Adaptability, Time Management

EXPERIENCE

Adveccio Pvt Ltd, Software Engineer Oct 2024 – Current

- Led the end-to-end development of Business Management System for VK Enterprises PVT LTD, covering business analysis, development, and deployment using **Flask, React, MySQL, Docker, CI/CD**
- Contributed to the development of a web-based ERP system for CBL Product Distribution System. **React, Flask, MySQL, Docker, CI/CD, Azure**
- AI-powered customer support Chatbot development for the SFA ERP system using **Python, RAG, LLaMA, LangChain, FAISS and FastAPI**
- Raw material consumption prediction system for a manufacturing company based on historical data. **Python, Flask, React, SARIMA, LSTM, MS SQL**
- Virtual Try-On application using NLP models from **Hugging Face** for garment and body image processing, for garment visualization (**React, Flask**)
- Contribute to the development of a SFA ERP System, delivering both web (**Flask, React, MySQL**) and mobile (**React Native**) applications to streamline order processing and distribution tracking.

Freelance Data Scientist – Fiverr Dec 2023 – Jul 2024

- Completed freelance projects on Fiverr analyzing e-commerce sales datasets to identify seasonal demand patterns and product performance using Python (**Pandas, NumPy**).
- Developed sales forecasting models (**Linear Regression, Random Forest**) for small retail businesses to predict monthly product demand and support inventory planning.
- Performed customer segmentation using **K-Means clustering** on online store transaction data to group customers based on purchasing behavior.
- Built data visualizations and analytical reports using **Matplotlib and Seaborn** to highlight product trends, revenue patterns, and customer purchasing frequency.

VEGA Innovations Pvt Ltd, Trainee (Battery Research and Development Section) Jun 2023 – Nov 2023

- Real-time CAN Data Logger Development (Group) - STM32 MCU Programming, PCB Designing
- CAN Data Decoding Algorithm Development (Individual) - **Python, DSA, NumPy, Pandas**

EDUCATION

MSc in Data Science and Artificial Intelligence

2026 – 2027

Department of Computer Science & Engineering, University of Moratuwa, Sri Lanka

Part-time Weekend Program

- **Advanced level coursework:** AI, ML, NLP, Data Science, Big Data Analytics, Databases, Data Mining

BSc (Hons) In Electrical and Electronic Engineering, University of Peradeniya, Sri Lanka

Nov 2019 – Aug 2024

Washington Accord Recognized

- **Relevant coursework:** Machine Learning, Image and Video Coding, Optimization, Industrial Statistics, Operations Research

PROJECTS

Improving Vehicle Number Plate Visibility via Low-Resolution Video Enhancement (Group)

- Develop an automated multi-model algorithm to enhance image visibility by integrating Image Processing and Super Resolution techniques (SRCNN and ESRGAN), Image Restoration, Motion Deblurring, and Tilt Correction **Python, NumPy, OpenCv, Tensorflow, Pytorch, Jupyter**

AI-Driven Defect Detection of Flawless Ceramic Products (Group)

- Develop a real-time YOLOv8-based model for detecting and classifying defects (cracks, iron spots, and pinholes) in ceramic products manufactured at Noritake Company. Data collection, annotation, image processing, training, and testing using **NVIDIA Jetson, YOLOv8, Python, NumPy, OpenCv**

Compliance Document Intelligence — RAG Service

- Built a RAG system for AML/KYC compliance documents enabling citation-backed, grounded answers from policy queries using LLMs. Hybrid retrieval (**dense embeddings + BM25 with RRF**), **FAISS** vector store, **BAAI/bge-small-en-v1.5** embeddings, **Groq LLaMA** with **Ollama LLaMA** fallback, **Pydantic v2** validation.

Image and Video Compression System (Individual)

- Develop a hybrid video codec incorporating DCT, quantization, prediction, entropy coding, and decoding featuring block-based motion prediction and optimized for both fixed and flexible bandwidth transmission **Python, NumPy, OpenCV**

Designing Treadmill Dashboard

- Develop a Python-based GUI application that displays real-time workout metrics, including speed, distance, calories burned, and step rate.

IoT-based Smart Medicine Pill Dispenser (Group)

- Dispensing medication according to a pre-set schedule, providing reminders and alerts, and control using wifi via a mobile app. (Arduino, ESP 32, Google Firebase)

Hospital management system for COVID-19 (individual)

- Develop a C program for patient registration, counting patients, and managing COVID-19 hospital operations.

CERTIFICATES

AWS Cloud Practitioner Essentials by Amazon web Service - Coursera 2024

AWS Educate Badges by Amazon Web Service

- AWS Academy Data Engineering, Getting Started with Compute, Getting Started with Serverless, Machine Learning Foundations

Generative AI with Large Language Models by DeepLearning.ai, AWS - Coursera 2025

Machine learning specialization by deeplearning.ai - Coursera 2023

- Supervised Machine Learning: Regression and Classification, Advanced Learning Algorithms, Unsupervised Learning, Recommenders, Reinforcement Learning

Neural Networks and Deep Learning by deeplearning.ai - Coursera 2024

DevOps Beginners to Advanced with Projects - Udemy 2024 (Ongoing)

- CI/CD, AWS, Linux, Scripts, Jenkins, Ansible, GitOps, Docker, Kubernetes, Terraform, & ChatGPT

Introduction to Cloud Computing by IBM - Coursera 2024

Programming for Everybody , Python Data Structures by University of Michigan - Coursera 2020